

Ethics Statement

This study is a retrospective secondary analysis of de-identified clinical imaging and structured records. Data access and use were reviewed and authorised through the institutional data-governance process of Henan Provincial People’s Hospital. No animal experiments, prospective enrolment, research-directed intervention, or identifiable participant information were involved. A formal ethics-committee approval number and consent or waiver determination are not available for reporting; this statement describes the available institutional data-governance authorisation.

Public benchmarks (BOP Linemod/T-LESS, SCARED, real-colon FEM, EndoNeRF, Blended-MVS) are used under their original providers’ access terms; these contain no new human or animal material acquired for this study. The clinical CT-to-bronchoscopy evaluation uses de-identified patient CT and intra-operative video provided by Henan Provincial People’s Hospital under the authorisation described above.